AI-DRIVEN MAINTENANCE REPORT

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1. Technical Sensor Data

Parameter	Measured Value
Air Temperature	300.0 °F
Process Temperature	340.0 °F
Rotational Speed	1200 RPM
Torque	65.0 Units
Tool Wear	210 %

2. AI Agent Analysis

Agent 1 (Initial Analysis):
Torque > 60 and Tool wear > 200 indicating high risk.

Agent 2 (Verification):
Torque > 60 and Wear > 200 trigger high risk rules.

3. Engineering Analysis & Recommendations

Summary:
REQUIRED MAINTENANCE

Analysis:
The sensor data indicates that the system is experiencing high-risk conditions. Specifically, the torque value of 65.0 exceeds the safety threshold of 60, indicating "high torque". Additionally, the tool wear value of 210 exceeds the threshold of 200, indicating "tool wear high". Furthermore, the air temperature of 300.0 and process temperature of 340.0 indicate an abnormal temperature difference of 40.0, exceeding the threshold of 25.

Validation:
The sensor values exceed the safety thresholds as follows:
* Torque > 60: Yes (65.0 > 60)
* Tool Wear > 200: Yes (210 > 200)
* Temperature Difference > 25: Yes (40.0 > 25)

Recommendation:
1. IMMEDIATELY reduce rotational speed to minimize torque and prevent further tool wear.
2. SCHEDULE an urgent maintenance session to inspect and replace the worn-out tools, as excessive tool wear can lead to system failure.
3. MONITOR temperature readings closely and take corrective action if temperatures continue to rise above normal levels, as high temperatures can cause thermal damage to system components.

4. Conclusion

Final Decision: REQUIRED MAINTENANCE
Decision Source: RULE-BASED SYSTEM TRIGGERED